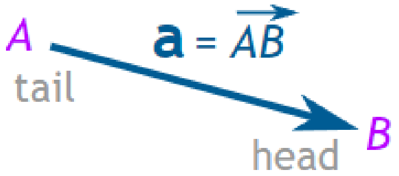
Vectors

Vectors

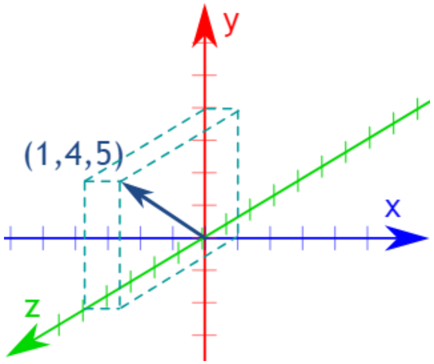
Vectors (velocity, force, acceleration, etc.) have both magnitude and direction.

A vector is often written in **bold**, like **a** or **b**.

A vector can also be written as the letters of its head and tail with an arrow above it, like this:



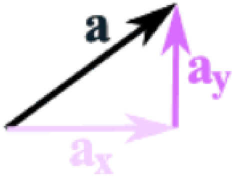
Vectors work perfectly well in 3 or more dimensions:



Vectors

Components of a Vector

The most common way is to first break up vectors into x and y parts, like this:



The vector **a** is broken up into
the two vectors $\mathbf{a_x}$ and $\mathbf{a_y}$

Magnitudes and direction

The “magnitude” of a vector is the distance from the endpoint of the vector to the origin – in a word, it’s length. Suppose we want to calculate the magnitude of the vector $\vec{a}$ = [4,3].

This vector extends 4 units along the x-axis, and 3 units along the y-axis.

$$\|\vec{a}\| = \sqrt{x^2 + y^2} = \sqrt{4^2 + 3^2} = 5$$

The magnitude of a vector is a scalar value – a number representing the length of the vector independent of the direction. There are a lot of examples were the magnitudes of vectors are important to us: velocities are vectors, speeds are their magnitudes; displacements are vectors, distances are their magnitudes.

To complement the magnitude, which represents the length independent of direction, one might wish for a way of representing the direction of a vector independent of its length. For this purpose, we use “unit vectors,” which are quite simply vectors with a magnitude of 1. A unit vector is denoted by a small “carrot” or “hat” above the symbol. For example, $\hat{a}$ represents the unit vector associated with the vector $\vec{a}$. To calculate the unit vector associated with a particular vector, we take the original vector and divide it by its magnitude. In mathematical terms, this process is written as:

$$\hat{a} = \frac{\vec{a}}{||\vec{a}||}$$

A unit vector is a vector of magnitude 1. Unit vectors can be used to express the direction of a vector independent of its magnitude.

$$\hat{a} = \frac{[4,3]}{5}$$

$$\hat{a} = \left[\frac{4}{5}, \frac{3}{5}\right]$$

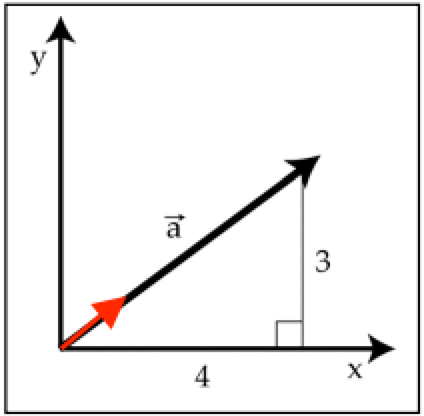
As shown in red in the figure, by dividing each component of the vector by the same number, we leave the direction of the vector unchanged, while we change the magnitude. If we have done this correctly, then the magnitude of the unit vector must be equal to 1 (otherwise it would not be a unit vector). We can verify this by calculating the magnitude of the unit vector $||\hat{a}||$

$$||\hat{a}||^2 = \left(\frac{4}{5}\right)^2 + \left(\frac{3}{5}\right)^2$$

$$||\hat{a}||^2 = \left(\frac{16}{25}\right) + \left(\frac{9}{25}\right)$$

$$||\hat{a}||^2 = \left(\frac{25}{25}\right) = 1$$

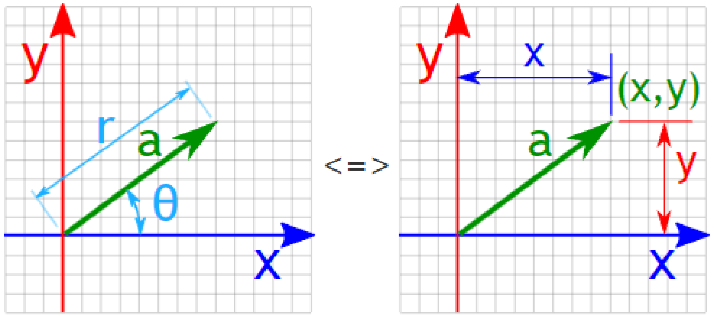
$$||\hat{a}|| = 1$$



Vectors

Vector Coordinates

We can calculate a vector's magnitude and direction, from its position (x and y length) and vice versa.



Vector **a** in Polar Coordinates

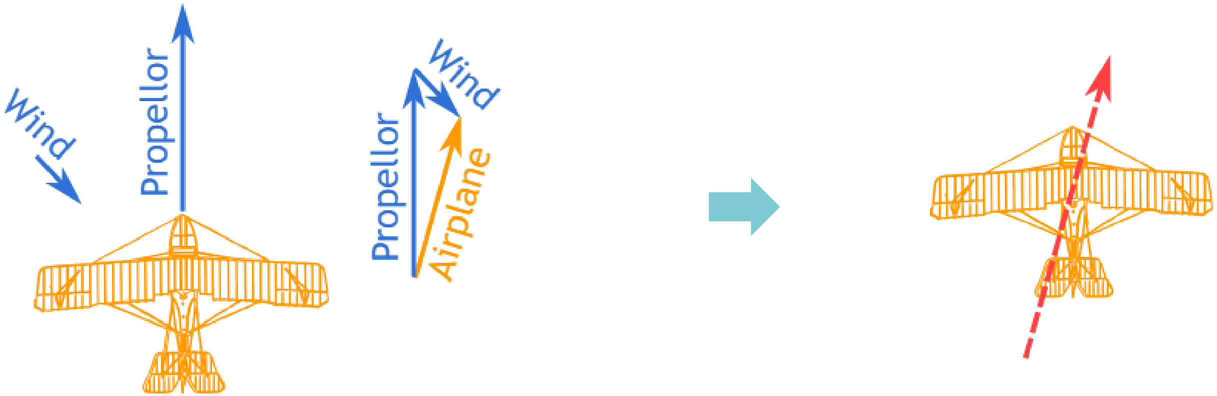
Vector **a** in Cartesian Coordinates

Polar (r,θ) >> Cartesian (x, y)	Cartesian (x, y) >> Polar (r,θ)
$x = r \times \cos\theta$ $y = r \times \sin\theta$	$r = \sqrt{(x^2 + y^2)}$ $\theta = \tan^{-1}(y / x)$

Vectors

Example

A plane is flying along, pointing North, but there is a wind coming from the North-West.



The two vectors (the velocity caused by the propeller, and the velocity of the wind) result in a slightly slower ground speed heading a little East of North. If you watched the plane from the ground, it would seem to be slipping sideways a little.

Vector addition and subtraction

Vectors can be added and subtracted. Graphically, we can think of adding two vectors together as placing two-line segments end-to-end, maintaining distance and direction.

if $\vec{a} = [4,3]$ and $\vec{b} = [1,2]$,

$$\vec{a} + \vec{b} = \vec{c}$$

$$\vec{c} = \vec{a} + \vec{b}$$

$$\vec{c} = [4,3] + [1,2]$$

$$\vec{c} = [4+1, 3+2]$$

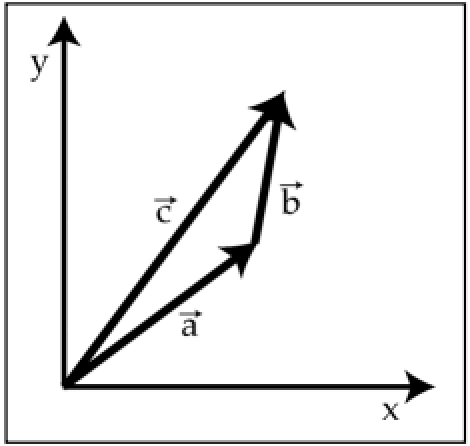
$$\vec{c} = [5,5]$$

Similarly, in vector subtraction:

$$\vec{c} = \vec{a} - \vec{b}$$

$$\vec{c} = [4,3] - [1,2]$$

$$\vec{c} = [3,1]$$



Vector addition has a very simple interpretation in the case of things like displacement. If in the morning a ship sailed 4 miles east and 3 miles north, and then in the afternoon it sailed a further 1 mile east and 2 miles north, what was the total displacement for the whole day? 5 miles east and 5 miles north – vector addition at work

Vector Space

All vectors live within a vector space.

A vector space is exactly what it sounds like – the space in which vectors live. When talking about spatial vectors, for instance, the direction and speed with which a person is walking through a room, the vector space is intuitively spatial since all available directions of motion can be plotted directly onto a spatial map of the room.

A vector space is a mathematical structure that consists of a set of vectors, along with operations of addition and scalar multiplication that satisfy specific properties.

Some important properties of vector space are:

- **Closure under Addition:** The sum of any two vectors in the vector space is also a vector in the vector space.
- **Closure under Scalar Multiplication:** Multiplying any vector in the vector space by a scalar yields another vector in the vector space.
- **Associativity of Addition:** Vector addition is associative, meaning $(u + v) + w = u + (v + w)$ for all vectors u, v, and w in the vector space.
- **Commutativity of Addition:** Vector addition is commutative, meaning $u + v = v + u$ for all vectors u and v in the vector space.
- **Existence of Additive Identity:** There exists a vector, denoted by 0 or 0, called the zero vector, such that $u + 0 = u$ for all vectors u in the vector space.
- **Existence of Additive Inverse:** For every vector u in the vector space, there exists a vector -u such that $u + (-u) = 0$.
- **Distributive Properties:** Scalar multiplication distributes over vector addition, meaning $\alpha(u + v) = \alpha u + \alpha v$ and $(\alpha+\beta)u = \alpha u + \beta u$ for all scalars α and β, and vectors u and v in the vector space.
- **Multiplicative Identity:** Scalar 1 acts as the multiplicative identity, meaning $1 \cdot u = u$ for all vectors u in the vector space.

Applications of Vector Spaces

•**Data Representation:** In many machine learning algorithms, data is represented as vectors. For example, images can be represented as vectors of pixel values, text documents can be represented as vectors of word counts or embeddings, and numerical data can be directly represented as vectors.

•**Feature Vectors:** Feature engineering involves creating meaningful representations of data. These representations are often in the form of feature vectors, where each feature corresponds to a dimension in the vector space. Feature vectors are used as input to machine learning models.

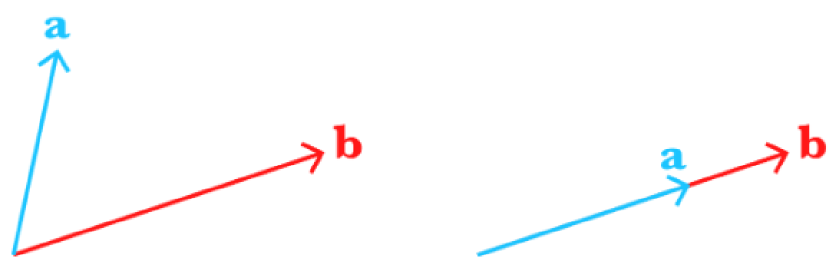
•**Vector Operations:** Vector operations such as addition, subtraction, dot products, and vector norms are commonly used in machine learning algorithms. For example, in clustering algorithms like k-means, vector addition and subtraction are used to calculate centroids.

•**Linear Algebra in Models**: Many machine learning models are based on linear algebra operations. For example, linear regression involves finding a line that best fits a set of data points, which can be formulated as a linear algebra problem involving vectors and matrices.

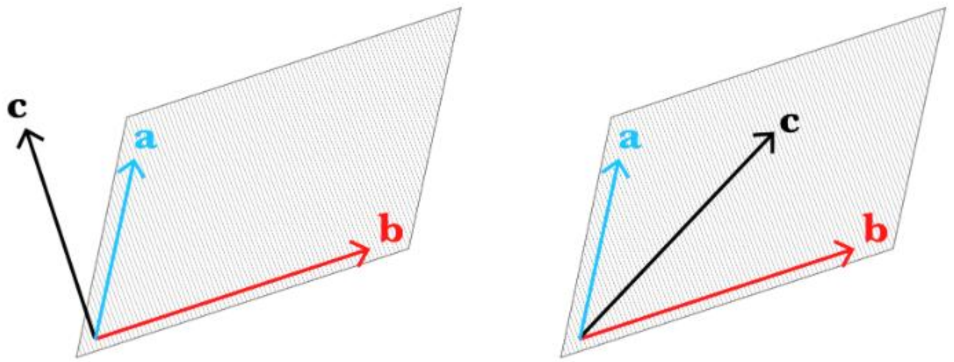
Linear independence

If two vectors point in different directions, even if they are not very different directions, then the two vectors are said to be linearly independent. If vectors $\vec{a}$ and $\vec{b}$ point in the same direction, then you can multiply vector $\vec{a}$ by a constant, scalar value and get vector $\vec{b}$, and vice versa to get from $\vec{b}$ to $\vec{a}$. If the two vectors point in different directions, then this is not possible to make one out of the other because multiplying a vector by a scalar will never change the direction of the vector, it will only change the magnitude.

A family of vectors is linearly independent if no one of the vectors can be created by any linear combination of the other vectors in the family.



Visualization of two linear independent (left) and dependent (right) vectors in 2-dimensions.



Visualization of three linear independent (left) and dependent (right) vectors in 3-dimensions.

Vector multiplication

Next, we move into the world of vector multiplication. There are two principal ways of multiplying vectors, called dot products and cross products.

$$d = \vec{a} \cdot \vec{b}$$

generates a scalar value from the product of two vectors. Do not confuse the dot product with the cross product:

$$\vec{c} = \vec{a} \times \vec{b}$$

$$\vec{a} \cdot \vec{b} = [4,3] \cdot [1,2]$$

$$\vec{a} \cdot \vec{b} = (4 * 1) + (3 * 2)$$

$$\vec{a} \cdot \vec{b} = 11$$

Another way of calculating the dot product of two vectors is to use a geometric means. The dot product can be expressed geometrically as:

$$\vec{a} \cdot \vec{b} = \|\vec{a}\| \left\|\vec{b}\right\| \cos\theta$$

Definition: A dot product (or scalar product) is the numerical product of the lengths of two vectors, multiplied by the cosine of the angle between them

where θ represents the angle between the two vectors.

Orthonormality

As the angle between the two vectors opens up to approach 90^0, the dot product of the two vectors will approach 0, regardless of the vector magnitudes

In this case, the two vectors are said to be orthogonal.

Two vectors are orthogonal to one another if the dot product of those two vectors is equal to zero.

A vector is a mathematical object that has both magnitude and direction, while a vector space is a mathematical structure consisting of a set of vectors along with operations of addition and scalar multiplication, satisfying specific properties. Vectors are elements of vector spaces, providing the algebraic framework for studying linear relationships and operations.

The Cross Product

If $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$, then the **cross product** of **a** and **b** is the vector

$$\mathbf{a} \times \mathbf{b} = \langle a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1 \rangle$$

- Notice that the **cross product a × b** of two vectors **a** and **b**, unlike the dot product, is a vector. For this reason, it is also called the **vector product**.

- Note that **a × b** is defined only when **a** and **b** are *three-dimensional* vectors.

The Cross Product

- If we now rewrite the definition using second-order determinants and the standard basis vectors **i**, **j**, and **k**, we see that the cross-product of the vectors
- $\mathbf{a} = a_1\,\mathbf{i} + a_2\,\mathbf{j} + a_3\,\mathbf{k}$ and $\mathbf{b} = b_1\,\mathbf{i} + b_2\,\mathbf{j} + b_3\,\mathbf{k}$ is

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \mathbf{k}$$

Example

- If **a** = ⟨1, 3, 4⟩ and **b** = ⟨2, 7, –5⟩, then

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 3 & 4 \\ 2 & 7 & -5 \end{vmatrix}$$

$$= \begin{vmatrix} 3 & 4 \\ 7 & -5 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 1 & 4 \\ 2 & -5 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 1 & 3 \\ 2 & 7 \end{vmatrix} \mathbf{k}$$

- $= (-15 - 28)\mathbf{i} - (-5 - 8)\mathbf{j} + (7 - 6)\mathbf{k}$
- $= -43\mathbf{i} + 13\mathbf{j} + \mathbf{k}$